

CIS 301: Logical Foundations of Programming
Spring 2026

Exam 1 – 100 points

This test is closed-notes and closed-computers.

This exam has 10 multiple choice and 5 open-ended problems.

Name: _____ Solution _____

Score: _____

Multiple choice: 10 questions, 3 points each.

Clearly write the letter of your answer next to each question.

1. On the island of knights and knaves, knights always tell the truth and knaves always lie. You meet two inhabitants of that island: Kirsten and Sven.

Kirsten says: "Sven would say that I am a knave."

Sven says: "Kirsten isn't a knave."

What type is Kirsten and what type is Sven?

- a) Kirsten is a knight and Sven is a knight
 - b) Kirsten is a knight and Sven is a knave
 - c) Kirsten is a knave and Sven is a knight
 - d) Kirsten is a knave and Sven is a knave**
2. Suppose we believe that two statements are logically equivalent. How can we prove this?
- a) Use truth tables to show that every truth assignment that makes the first statement true also makes the second statement true
 - b) Find a truth assignment where both statements have the same output
 - c) Use truth tables to show that the two statements have the same output for every possible truth assignment**
 - d) Use truth tables to show that the two statements have the same classification (tautology, contingent, or contradictory)
3. Consider the following statements:
- I will bake brownies if I get home before 6:00.
I don't bake brownies.

What can we conclude?

- a) I got home before 6:00
- b) I got home at 6:00 or later**
- c) We can't conclude anything.

4. Which of the following statements is logically equivalent to $p \wedge q \rightarrow r$?

- a) $r \rightarrow p \wedge q$
- b) $\neg r \rightarrow (\neg p \wedge \neg q)$
- c) $\neg p \vee \neg q \vee \neg r$
- d) $(\neg p \wedge \neg q) \vee r$

NO SOLUTION. The correct solution was supposed to be $\neg p \vee \neg q \vee r$, but there was a typo on C. Everyone got points for this question.

5. Suppose we know that a logical statement is satisfiable. What can we conclude?

- a) The statement is a tautology
- b) The statement is contingent
- c) We can't find a truth assignment where the statement outputs false
- d) **None of the above**

6. Consider the following argument. What truth assignment demonstrates that the argument is invalid?

Premises: $p \vee q, \neg r \rightarrow p, q \rightarrow \neg p$

Conclusion: $r \rightarrow q$

- a) $p = T, q = T, r = F$
- b) **$p = T, q = F, r = T$**
- c) $p = F, q = T, r = F$
- d) $p = F, q = F, r = T$

7. Given the propositional atoms below, what is the best translation of the sentence, "I am not hungry or thirsty, but I am tired", to propositional logic?

p: I am hungry

q: I am thirsty

r: I am tired

- a) **$\neg(p \vee q) \wedge r$**
- b) $\neg p \vee \neg q \wedge r$
- c) $(\neg p \vee \neg q) \wedge r$
- d) $(\neg p \vee q) \wedge r$

8. What is the top-level operator in the logical statement below?

$$p \wedge q \vee \neg r \vee \neg s \wedge \neg q$$

- a) The rightmost $\wedge$
- b) The leftmost $\vee$
- c) The rightmost $\vee$**
- d) The leftmost $\wedge$

9. What is the negation of the condition in the following if-statement?

```
if (x > 7 && y != 14 || Character.isDigit(input)) {  
    //statements  
}
```

- a) `!(x > 7) && y == 14 && !Character.isDigit(input)`
- b) `(x < 7 || y == 14) && Character.isDigit(input)`
- c) `x <= 7 || (y == 14 && !Character.isDigit(input))`
- d) `(x <= 7 || y == 14) && !Character.isDigit(input)`**

10. Which of the following statements is contradictory?

- a) $(p \rightarrow q) \wedge (p \wedge \neg q)$**
- b) $\neg p \vee q \vee \neg p \vee q$
- c) $(p \vee q) \wedge (\neg p \vee \neg q)$
- d) $(p \rightarrow q) \vee (\neg p \wedge q)$

Open Ended: 5 questions, 9-20 points each.

11. (9 pts) Consider the following code fragment, where *marker*, *a*, *b*, and *c* are all *int* variables:

```
marker = 10;
if (a < 4 || b < 10 || c == 17) {
    marker = 1;
}
else if (a < 7 && c < 18) {
    if (a < 2 || b >= 5) {
        marker = 2;
    }
    else {
        marker = 3;
    }
}
// ← suppose we are right here
```

Suppose *marker* has the value 2 immediately after the above code fragment finishes (i.e., at the point marked with the ← just after the brackets). Answer the three questions below.

List the range of values that are possible for *a*.

a can be 4, 5, or 6

List the range of values that are possible for *b*.

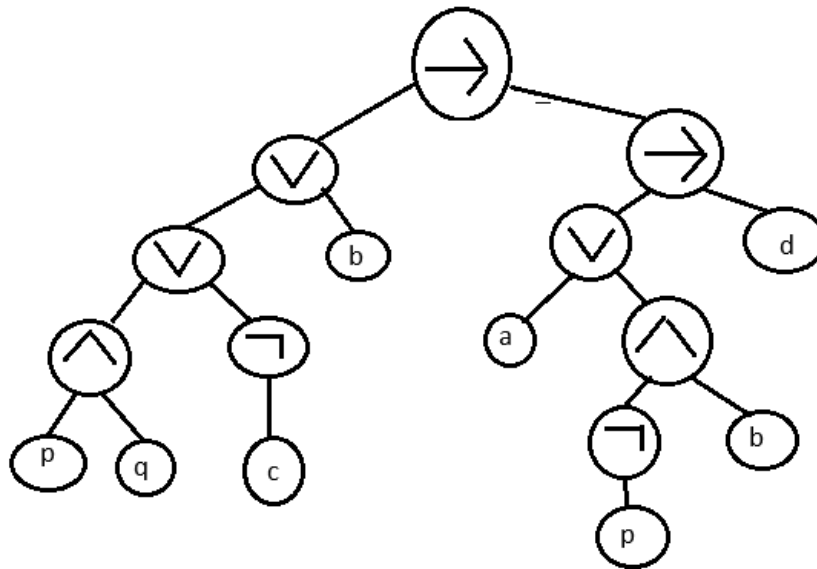
b must be >= 10

List the range of values that are possible for *c*.

c < 17

12. (11 pts) Draw a parse tree for the following logical formula:

$$p \wedge q \vee \neg c \vee b \rightarrow a \vee \neg p \wedge b \rightarrow d$$



13. (12 pts) Consider the natural deduction proof below. Next to each ???, fill in the missing justification. (You can assume that “SomeJustification” is a valid justification for the given proof step – you only need to fill in each ???.)

- | | |
|---------------------|--------------------------|
| 1 (a ∧ b) | by SomeJustification, |
| 2 (a) | by AndE1(1) , |
| 3 (a ∨ p ∧ q) | by OrI1(2) , |
| 4 (b ∨ d ∨ e → q) | by SomeJustification, |
| 5 (q → a ∧ p) | by SomeJustification, |
| 6 (b) | by AndE2(1) , |
| 7 (b ∨ d) | by OrI1(6) , |
| 8 (b ∨ d ∨ e) | by OrI1(7) , |
| 9 (q) | by ImplyE(4, 8) , |
| 10 (a ∧ p) | by ImplyE(5, 9) |

14. (18 pts) Consider the blank combined truth table below for a proposed argument.

		*			*			*			*		

p	q	r	#	(	p	→	r,	¬q	→	p,	r ∨ ¬q) ⊢ ¬p ∨ q		

T	T	T	#		T		F	T		T	F	F	T
T	T	F	#		F		F	T		F	F	F	T
T	F	T	#		T		T	T		T	T	F	F
T	F	F	#		F		T	T		T	T	F	F
F	T	T	#		T		F	T		T	F	T	T
F	T	F	#		T		F	T		F	F	T	T
F	F	T	#		T		T	F		T	T	T	T
F	F	F	#		T		T	F		T	T	T	T

- a) (5 pts) Complete the truth table by filling in the output under each operator. Use a * to mark the top-level operator for each premise and the conclusion.
- b) (4 pts) Circle one option: is the argument above valid or **invalid**?

Valid

Invalid

- c) (4 pts) If you circled "**valid**" above, define what it means for an argument in propositional logic to be valid. If you circled "**invalid**" above, define what it means for an argument in propositional logic to be invalid.

An argument is invalid if we can find a truth assignment where all the premises are true but the conclusion is false.

- d) (5 pts) Explain your reasoning for part (b) by citing specific relevant truth assignment(s) from your truth table and using the definition of a valid or invalid argument.

For the truth assignment $p = T, q = F, r = T$, all the premises are true but the conclusion is false.

15. (20 pts) Complete the following natural deduction proof

$(a \vee (b \wedge (p \vee q))) \vdash ((a \vee b) \wedge (p \vee q \vee a))$

Proof(

1 ($a \vee (b \wedge (p \vee q))$)

by Premise,

//FINISH THE PROOF BELOW

2 SubProof(

3 Assume (a),

4 ($a \vee b$)

by OrI1(3),

5 ($p \vee q \vee a$)

by OrI2(3),

6 ($(a \vee b) \wedge (p \vee q \vee a)$) by AndI(4, 5)

),

7 SubProof(

8 Assume ($b \wedge (p \vee q)$),

9 (b)

by AndE1(8),

10 ($p \vee q$)

by AndE2(8),

11 ($a \vee b$)

by OrI2(9),

12 ($p \vee q \vee a$)

by OrI1(10),

13 ($(a \vee b) \wedge (p \vee q \vee a)$)

by AndI(11,12)

),

14 ($(a \vee b) \wedge (p \vee q \vee a)$)

by OrE(1,2,7)

)